Case Study Project #2 Proposal Revision Notes

Full Capacity: The Evolution of Computer Data Storage

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Section 1: Overview of Revisions

This document summarizes the revisions made to the group's project proposal following initial submission. Two major changes were applied: the expansion of key concept descriptions per exhibit section, and the addition of a mockup for the interactive element.

Section 2: Revision 1 - Key Concept Descriptions

Each exhibit section originally listed key concepts as labels only, with no accompanying explanation. In this revision, every bulleted key concept was expanded with a short declarative description to give reviewers a clearer understanding of what each topic covers. This was applied across all five exhibit sections: The Origin, The Disk, The Optical Age, The Flash, and The Horizon.

Table 1. Summary of Key Concept Revisions per Exhibit Section

Exhibit	Revision
The Origin: Punch Cards and Magnetic Drums	**<u>Before:</u>** Key concepts: ● Punch card data encoding and processing ● Batch Processing Systems ● Magnetic drum memory operations ● Sequential data access ● Early computer input and storage methods ● Foundations of digital data storage

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<td><u>After:</u>
Key concepts:

Punch card data encoding and processing
Explains how information was represented using holes punched into cards, allowing computers to read and process data and instructions.
Batch Processing Systems
Describes how jobs were grouped together and processed automatically in sequence without direct user interaction.
Magnetic drum memory operations
Covers how magnetic drums stored data on a rotating surface and how read/write heads accessed information as the drum spun.
Sequential data access
Explains how data was stored and retrieved in a fixed order, requiring earlier records to be read before reaching the desired data.
Early computer input and storage methods
Introduces the use of punch cards, paper tape, and magnetic drums as the primary methods for entering, storing, and retrieving data in early computers.
Foundations of digital data storage
Connects these early technologies to modern storage systems by showing how they established the basic principles of digital data encoding, storage, and retrieval.
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<td rowspan="2">The Disk: Magnetic Storage and HDDs</td>
<td><u>Before:</u>
Key Concepts:

Magnetic recording principles
Platters, read/write heads, and actuator arms
Random access vs. sequential access
Storage capability scaling and areal density
HDD role in memory hierarchy
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<td><u>After:</u>
Key Concepts:

Magnetic recording principles
Covers how data is stored and retrieved using magnetic fields, and how read/write heads interact with magnetized surfaces to encode binary information.
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	● Platters, read/write heads, and actuator arms ○ Describes the physical components of a hard disk drive, how they work together to access data, and how their design evolved over time. ● Random access vs. sequential access ○ Explains the difference between random and sequential access, and how HDDs improved upon earlier storage technologies by allowing data to be retrieved in any order. ● Storage capability scaling and areal density ○ Introduces how manufacturers increased storage capacity over time by packing more data into the same physical space through advances in areal density. ● HDD role in memory hierarchy ○ Connects hard disk drives to the broader memory hierarchy which describes how HDDs served as the primary form of secondary storage for decades alongside faster but smaller primary memory.
The Optical Age: CDs, DVDs, and Their Limits	**<u>Before:</u>** Key concepts: ● Optical data storage utilizing lasers ● CD and DVD data encoding and retrieval ● Pits and Lands as Binary Data Representation ● Digital Media Distribution and Archival Storage ● Offline/Tertiary Storage and Reimagination of Storage
	<u>After:</u> Key concepts: ● Optical data storage utilizing lasers ○ Covers how lasers are used to read and write data for optical data ○ Understand how the properties of light enable data to be saved from the reflective surfaces ● CD and DVD data encoding and retrieval ○ Describes how different types of media such as audio, video, and digital data are encoded onto CDs and DVDs ○ Know how the data reading process translates physical features into usable information digitally ● Pits and Lands as Binary Data Representation ○ Explains how the pits and flat lands on the disc represent

	binary 0s and 1s
	● Digital Media Distribution and Archival Storage
	  ○ Explores how discs became the preferred format for distributing media
	  ○ How their durability and standardization made them suitable for long-term archives
	● Offline/Tertiary Storage and Reimagination of Storage
	  ○ Connects optical media to the part it played in the storage hierarchy for offline storage
	  ○ Know how its limitations and strengths inspired more innovative ideas on how data could be preserved and accessed beyond primary and secondary memory

The Flash: SSDs, NAND, and NVMe	**Before:** Key concepts: ● NAND Flash Memory Architecture ● Solid-State Drives (SSDs) ● Non-Volatile Storage ● Wear Leveling and Flash Memory Lifespan ● SATA vs. NVMe Interfaces ● PCI Express (PCIe) Communication ● Storage Performance and Latency ● Input/Output Operations Per Second (IOPS) ● Modern Memory Hierarchy

	After:
	Key concepts:
	● NAND Flash Memory Architecture
	  ○ Explains the internal structure of NAND flash memory, including memory cells, pages, and blocks, and how electrical charges are used to represent binary data in non-volatile forms.
	● Solid-State Drives (SSDs)
	  ○ Describes how SSDs use flash memory chips, controllers, and firmware to store and manage data electronically, eliminating the mechanical limitations of traditional hard disk drives.
	● Non-Volatile Storage

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Covers the concept of storage that retains data even when power is removed, highlighting its importance in modern computing systems and long-term data preservation.

Wear Leveling and Flash Memory Lifespan

Introduces the limitations of NAND flash cells, which can only endure a finite number of write cycles, and explains how wear-leveling algorithms distribute writes evenly to extend drive longevity.

SATA vs. NVMe Interfaces

Compare the two major SSD interfaces, showing how SATA SSDs remain constrained by legacy hard drive communication standards while NVMe SSDs are designed to fully utilize the speed of flash storage.

PCI Express (PCIe) Communication

Explains how NVMe drives communicate directly through PCIe lanes, reducing protocol overhead and enabling much higher bandwidth than traditional storage interfaces.

Storage Performance and Latency

Examines the factors that affect storage speed, including access time, throughput, and latency, and how SSDs dramatically reduce delays compared to mechanical drives.

Input/Output Operations Per Second (IOPS)

Covers the measurement of storage responsiveness by evaluating how many read and write operations a device can perform within a second, a key metric in enterprise and high-performance computing environments.

Modern Memory Hierarchy

Connects SSDs and NVMe storage to the broader memory hierarchy, illustrating how advances in flash technology have narrowed the performance gap between secondary storage and primary memory while supporting the growing demands of modern applications, cloud computing, and data-intensive workloads.

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<td>The Horizon: Cloud, DNA, and Emerging Storage

Note: Merged some concepts</td>
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<u>Before:</u>
Key concepts:

Cloud Storage Architecture
Data Centers
Distributed Storage Systems
DNA Storage

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<tr><td rowspan="2">into one to explain related concepts better.</td><td>

- Modern Data Storage Methods
- Data Access and Retrieval
- Storage Capacity and Scalability
- Data Backup and Redundancy
- Emerging Storage Technologies

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<u>After:</u>

Key concepts:

- Cloud Storage Architecture
    - Explains how cloud storage works at a high level, including how data is sent, stored, and retrieved across remote servers through the internet.
- Data Centers
    - Describes what data centers are, how they are physically structured, and how they power and cool thousands of servers to keep data available 24/7.
- Distributed Storage Systems and Data Reliability
    - Introduces concepts such as RAID, data replication, sharding, backup, and redundancy, explaining how large-scale storage systems distribute and duplicate data across multiple servers and locations to improve reliability, availability, and performance.
- DNA Storage
    - Discusses how biological DNA can be used to encode digital data, its current state of research, its massive storage potential, and the challenges that still prevent real-world adoption.
- Cloud Storage and Scalability
    - Explains how cloud storage enables users and organizations to store and access data through remote servers while easily scaling storage capacity to meet growing data demands.
- Data Access and Retrieval
    - Explains how stored data is located, accessed, and retrieved efficiently through indexing, caching, networking, and storage management techniques.
- Emerging Storage Technologies
    - Briefly covers next-generation storage ideas such as holographic storage and neuromorphic storage, giving

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	visitors a glimpse of what the future of data storage may look like.

Section 3: Revision 2 - Interactive Element Mockup

UI mockups for each interactive element across the exhibit were added to the proposal to clarify the nature of the interactive components. These mockups demonstrate that each simulator and interactive feature is designed to respond to user input and provide dynamic visual feedback, rather than function as a static, click-through presentation. The mockups were added at Section IV. Tentative Style Guide Snapshot, labeled as Figures 3 to 7.

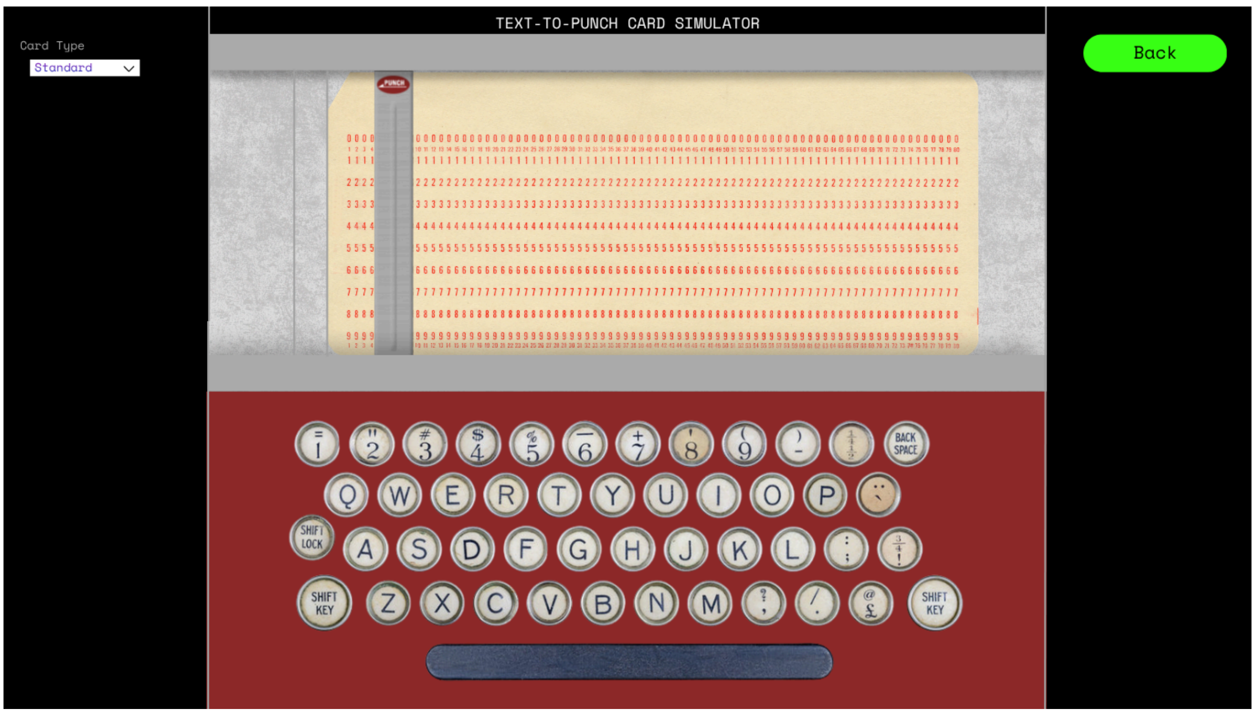

Figure 1. Mockup of Interactive Element 1: Text-to-Punch Card Simulator

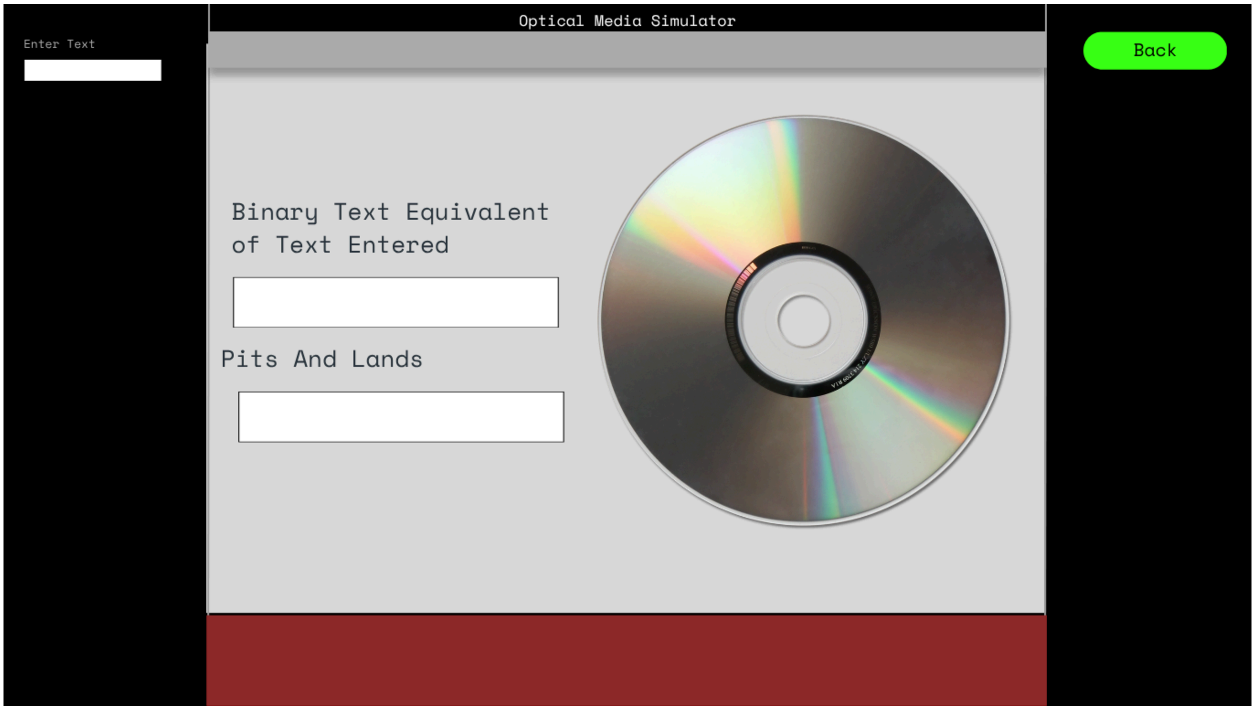

Figure 2. Mockup of Interactive Element 2: Optical Media Presentation

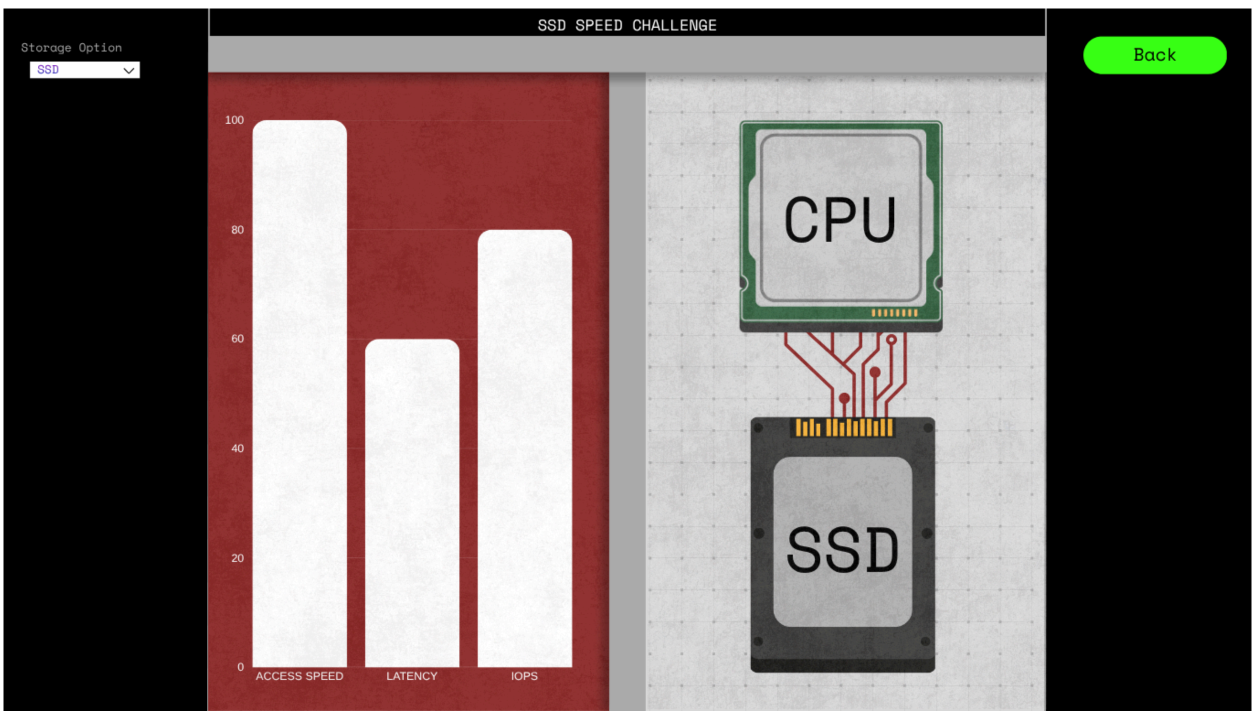

Figure 3. Mockup of Interactive Element 3: SSD Speed Challenge

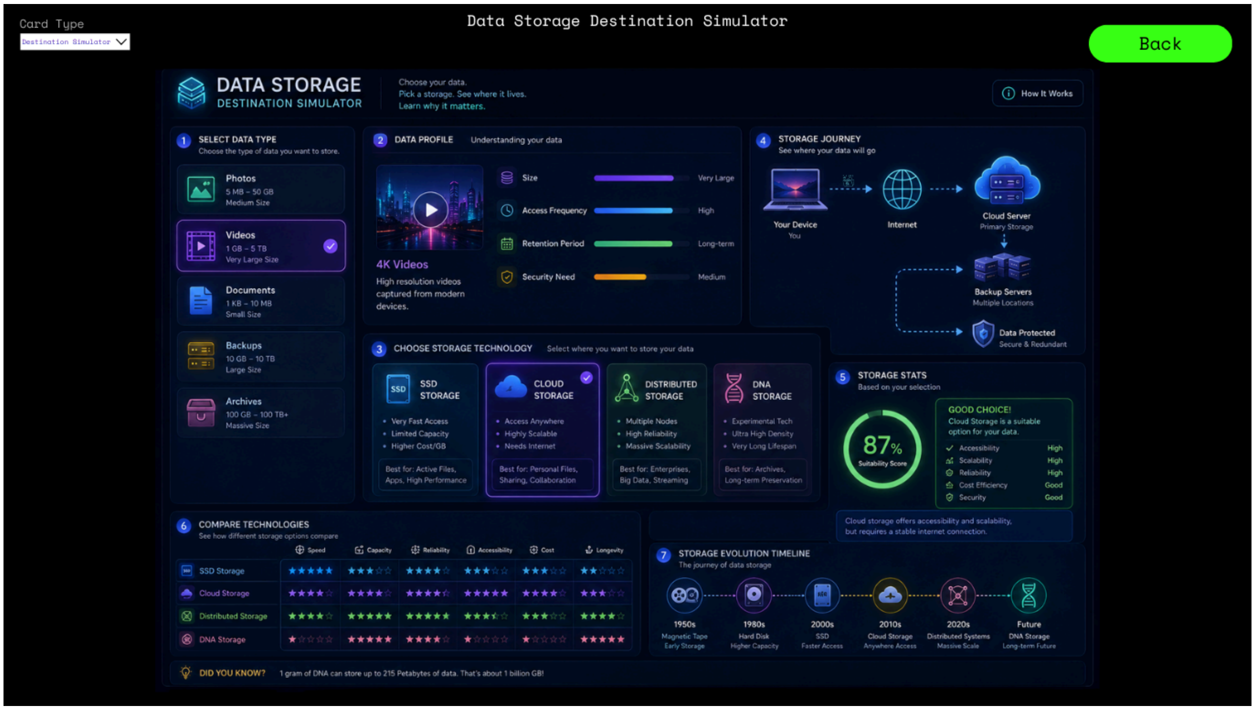

Figure 4. Mockup of Interactive Element 4: Data Storage Destination Simulator

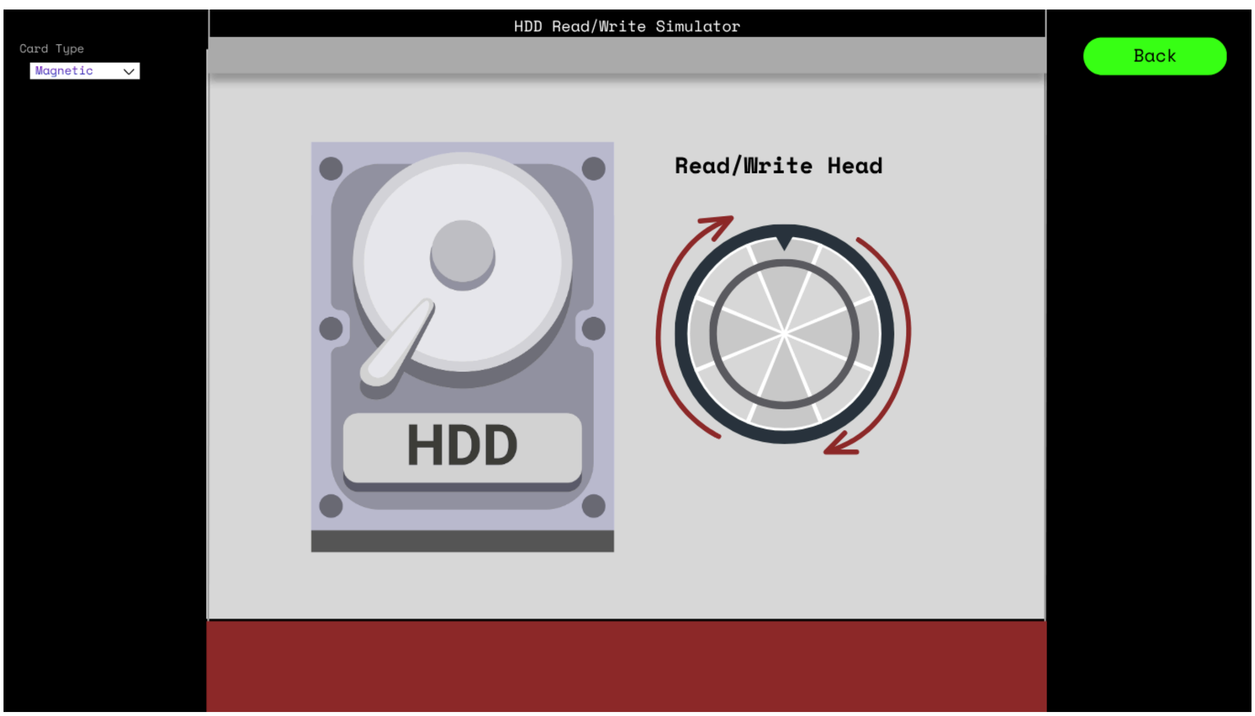

Figure 5. Mockup of Interactive Element 5: HDD Read/Write Simulator